

Question 01

$$\text{Demand Equation : } P = 800 - 10Q_d \quad \text{--- (1)}$$

$$\text{Supply Equation : } P = 5Q_s + 200 \quad \text{--- (2)}$$

a) At equilibrium point

$$Q_d = Q_s$$

① $\Rightarrow$

$$P = 800 - 10Q_d$$

$$Q_d = \frac{800 - P}{10} \quad \text{--- (A)}$$

② $\Rightarrow$

$$P = 5Q_s + 200$$

$$Q_s = \frac{P - 200}{5} \quad \text{--- (B)}$$

$$(A) = (B)$$

$$\frac{800 - P}{10} = \frac{P - 200}{5}$$

$$800 - P = 2P - 400$$

$$1200 = 3P$$

$$P = 400$$

$$P = 400$$

③ $\Rightarrow$

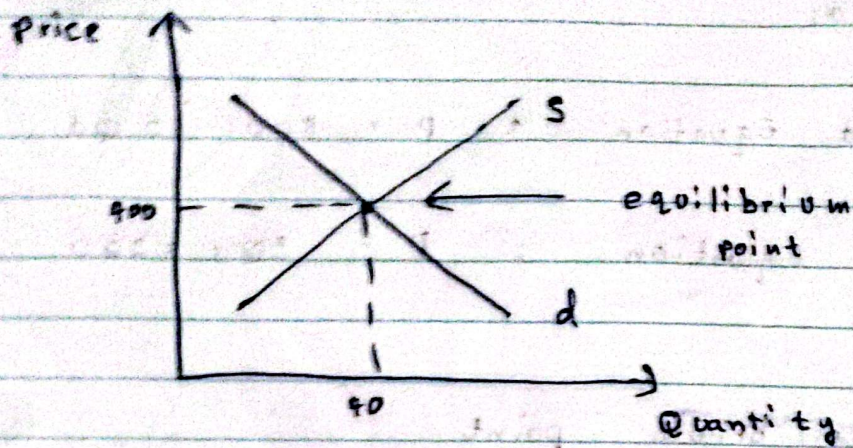
$$400 = 800 - 10Q_d$$

$$Q_d = \frac{400}{10}$$

$$Q_d = 40$$

Equilibrium price = LKR 400 Equilibrium quantity = 40

b)



c)

Price Cap = 400 (LKR)

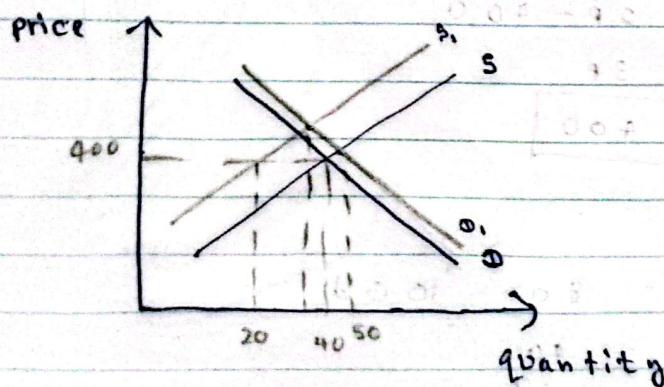
Equilibrium price = LKR 400

price cap is NOT binding

There is no shortage or no surplus, because price is same Equilibrium price. therefore market remain unchanged.

passengers (consumers) pay normal market price. Drivers (producer) receive normal earnings. Therefore platform operates efficiently.

d)



* Supply decreases by 20 - Supply curve shift left

* Demand increase by 10 - demand curve ~~increa~~ shift right.

Market price - increase.

both shifts ($\downarrow$ supply, $\uparrow$ demand)
push price upward.

Number of rides - decrease